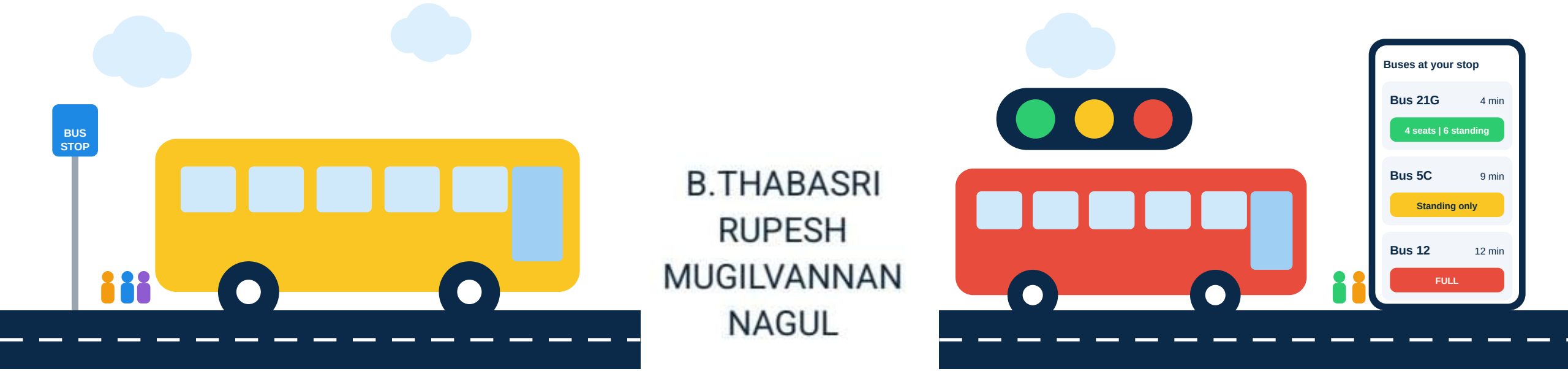
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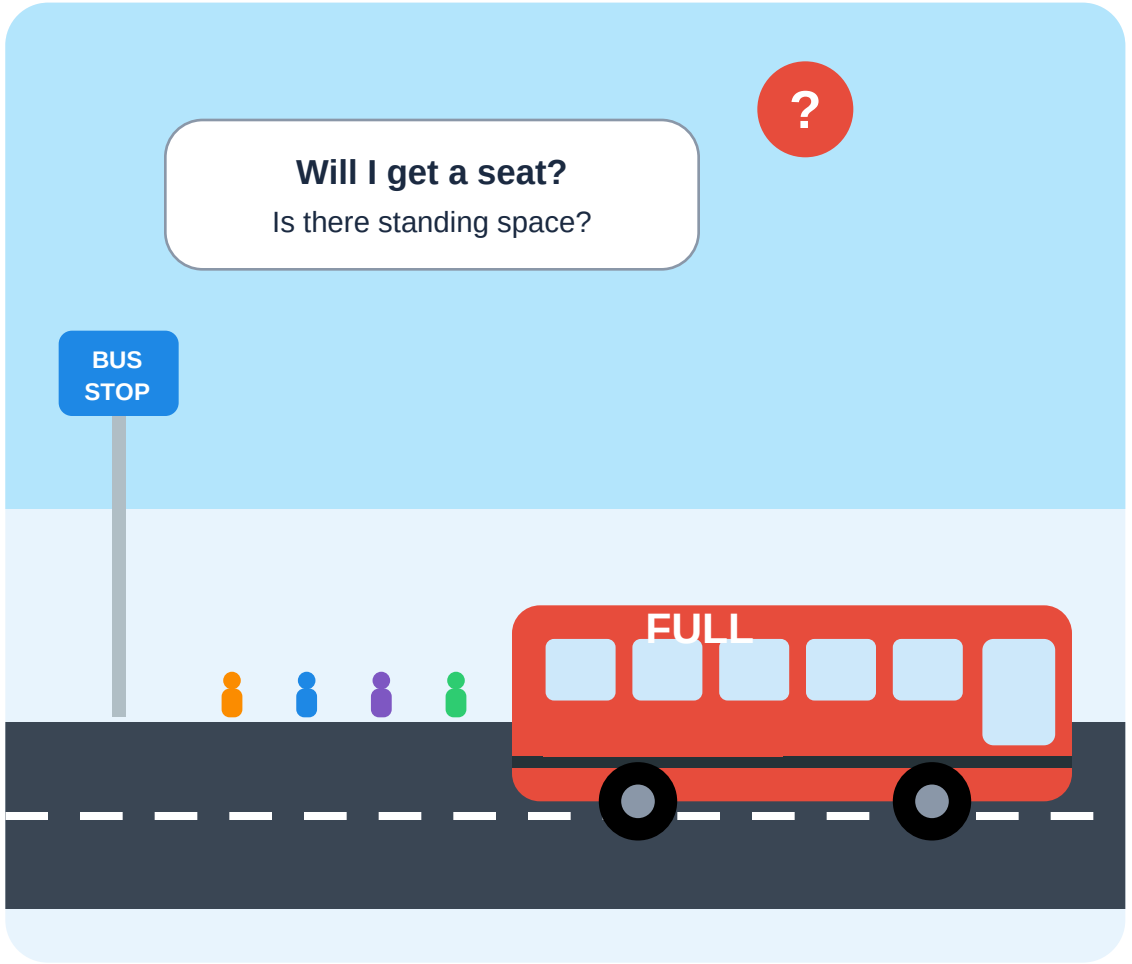
Next-Generation Bus Ticketing System with Intelligent Seat Reservation



B.THABASRI
RUPESH
MUGILVANNAN
NAGUL

The Problem

- People wait at the bus stop **without knowing** whether the next bus has a seat, standing space, or is totally full.
- The bus arrives, is **packed**, and passengers lose 10-20 minutes waiting for the next one.
- Many passengers are ready to **stand and travel** - they only want to know the situation **before** the bus comes.
- **Elders, women with children, students with bags** and office-goers all need to plan better.
- Seats booked for a short part of a journey stay **blocked** for the whole route.

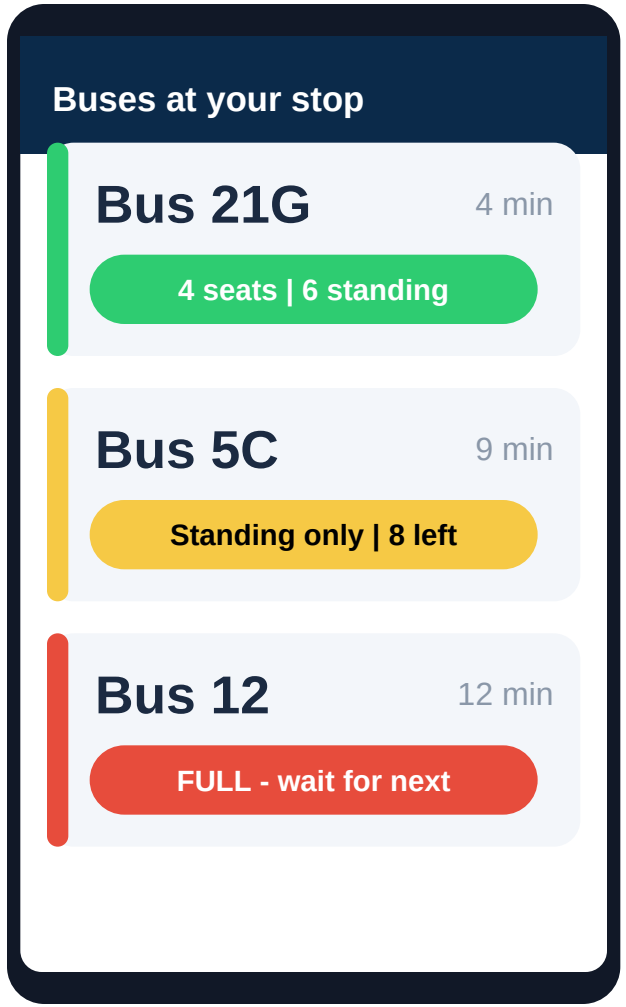


Core question: How can a person at the bus stop see the seat and standing-space situation of the *next bus* on their phone?

Our Solution

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See it on your phone



Seats available

You can sit. The app shows how many seats and standing places are free.

Standing only

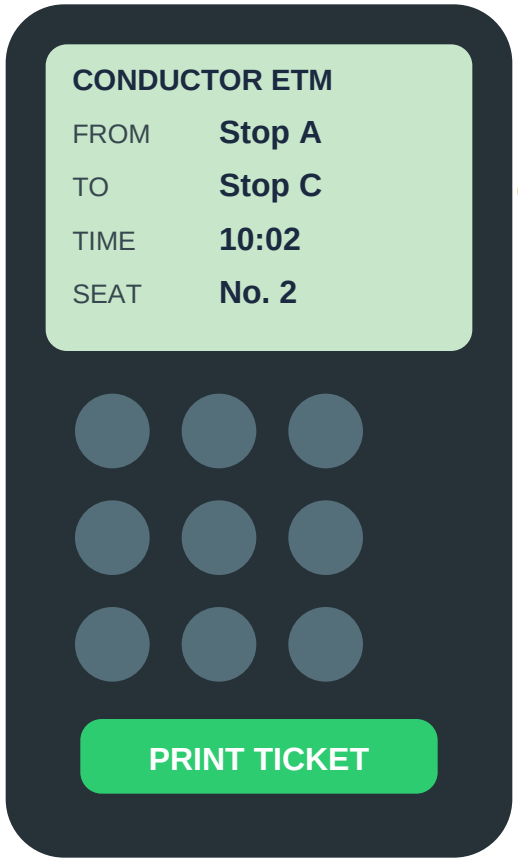
Seats are full but standing space is left. Many people are happy to stand - now they can decide.

Full

No space at all. Wait for the next bus instead of losing your time.

Live status - updated every few seconds - no personal data needed

Method 1: Tickets from the ETM



The tool conductors already use

BUS TICKET

Stop A to Stop C

Time: 10:02

Seat: 2

What one ticket tells the system

- **Boarding stop**
- **Destination stop**
- **Booking time**
- **Number of passengers**

Ticket issued = seat taken from boarding stop to destination

Destination reached = seat refreshes and is free again

Tickets booked = seats full; check on your phone

No new hardware on the bus - the ETM is already there

Method 2: Seats & Standing Space

No.	Time	From	To	Gets
1	10:00	Stop A	Stop C	Seat 1
2	10:02	Stop A	Stop D	Seat 2
3	10:04	Stop B	Stop E	Seat 3
4	10:05	Stop B	Stop C	Seat 4
5	10:07	Stop B	Stop D	Standing

Example bus: 4 seats + 2 standing places. Tickets are listed in booking order.

First come, first served rules

- Earlier booking time gets a seat first.
- As many tickets booked = as many seats full.
- When all 4 seats are taken, new tickets get standing places (maximum 2).
- When a seat is freed, the earliest standing ticket gets it.

After stop B

STANDING ONLY

Seats full (4 of 4). 1 of 2 standing places left.

After stop C

1 SEAT AVAILABLE

Tickets 1 and 4 ended. Ticket 5 moves to a seat. 2 standing free.

After stop D

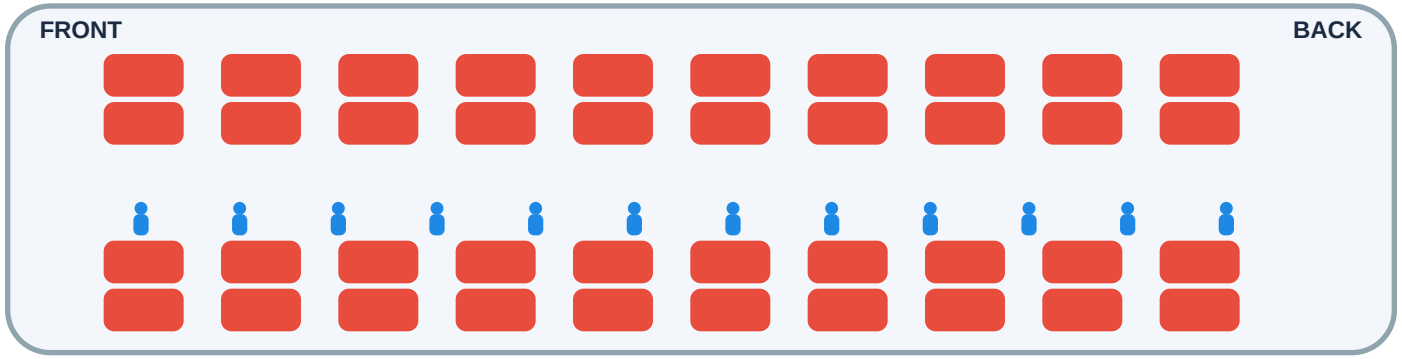
3 SEATS AVAILABLE

Tickets 2 and 5 ended. Only ticket 3 is on board.

Passengers see this status on their phone at the bus stop.

Method 2: Inside the Bus

BEFORE STOP C: 52 tickets on board = 40 seated + 12 standing



Stop C reached: 14 tickets end

AFTER STOP C: 38 on board = 38 seated, 2 seats free



Free seat Taken seat Standing person

Example bus: 40 seats, standing capacity 30. Seats are given first come, first served.

Data from ETM tickets

Total seats	**40**
Standing capacity	**30**
Tickets on board (before C)	**52**
Tickets ending at stop C	**14**

Calculation

On board after C = 52 - 14 = 38

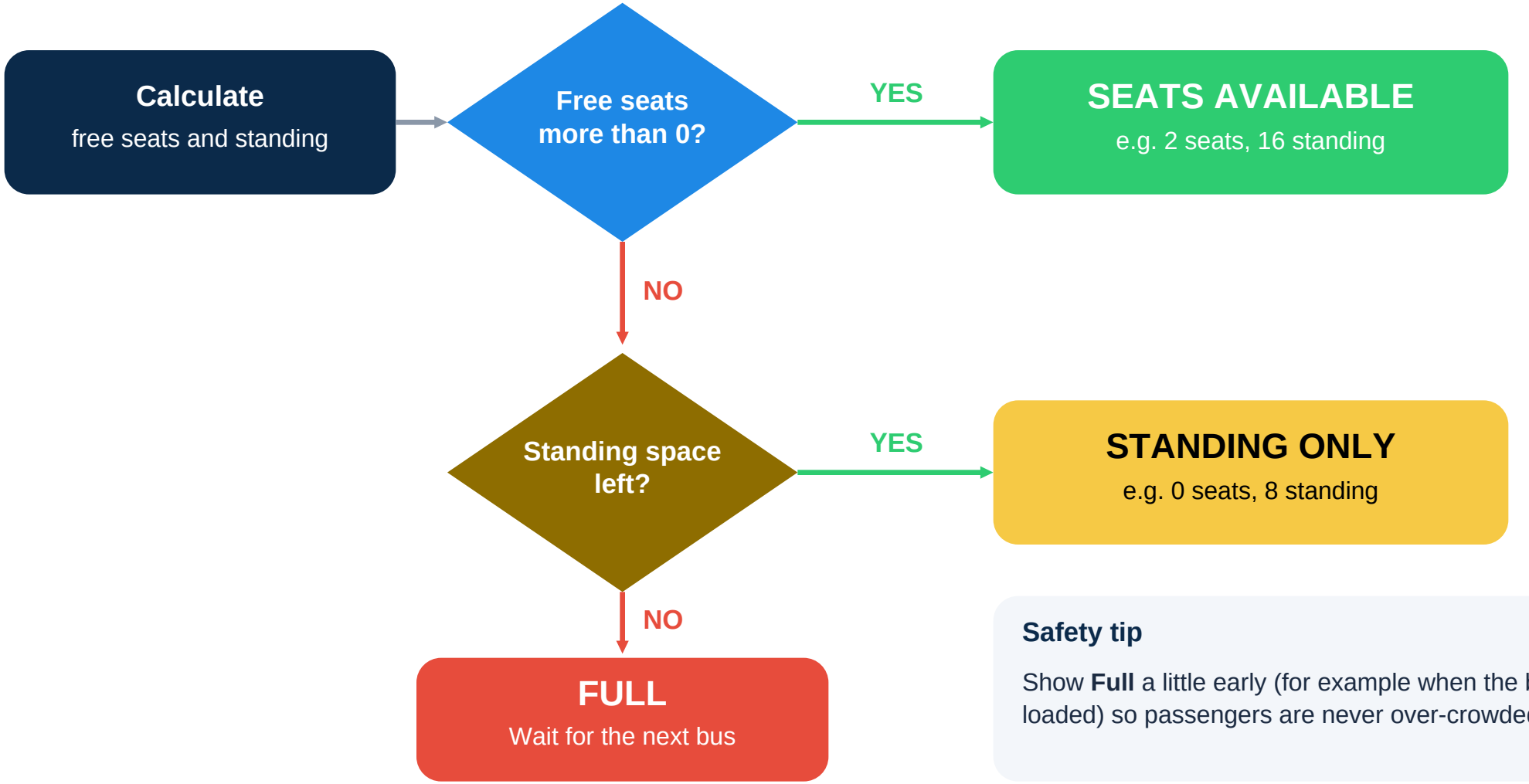
Free seats = 40 - 38 = 2

Free standing space = 30 - 0 = 30

App shows: 2 seats, 30 standing

1 ticket = 1 person on board. The first 40 tickets get seats; later tickets stand. When tickets end, standing passengers take the free seats.

Method 3: Deciding the Colour



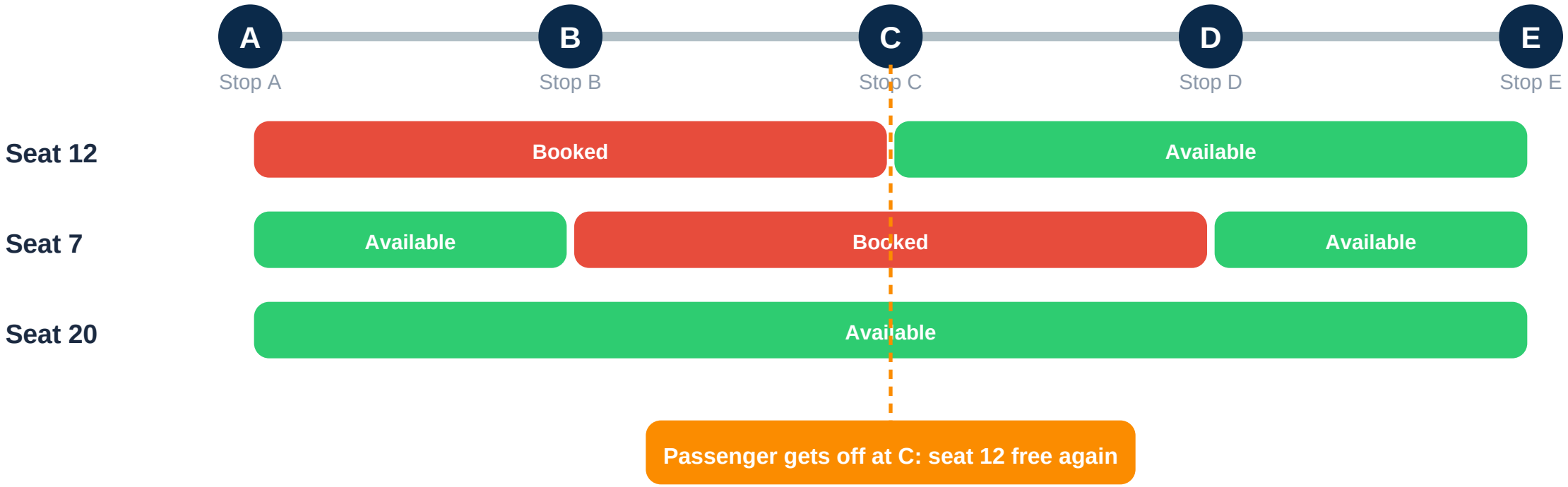
Safety tip

Show **Full** a little early (for example when the bus is about 90% loaded) so passengers are never over-crowded.

Method 4: Smart Seat Reservation

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Our unique feature



1 Part-journey booking

Seat 12 booked from stop A to stop C is blocked only between A and C.

2 Auto-free after stop C

After stop C the app shows seat 12 as available for people boarding at C, D or E.

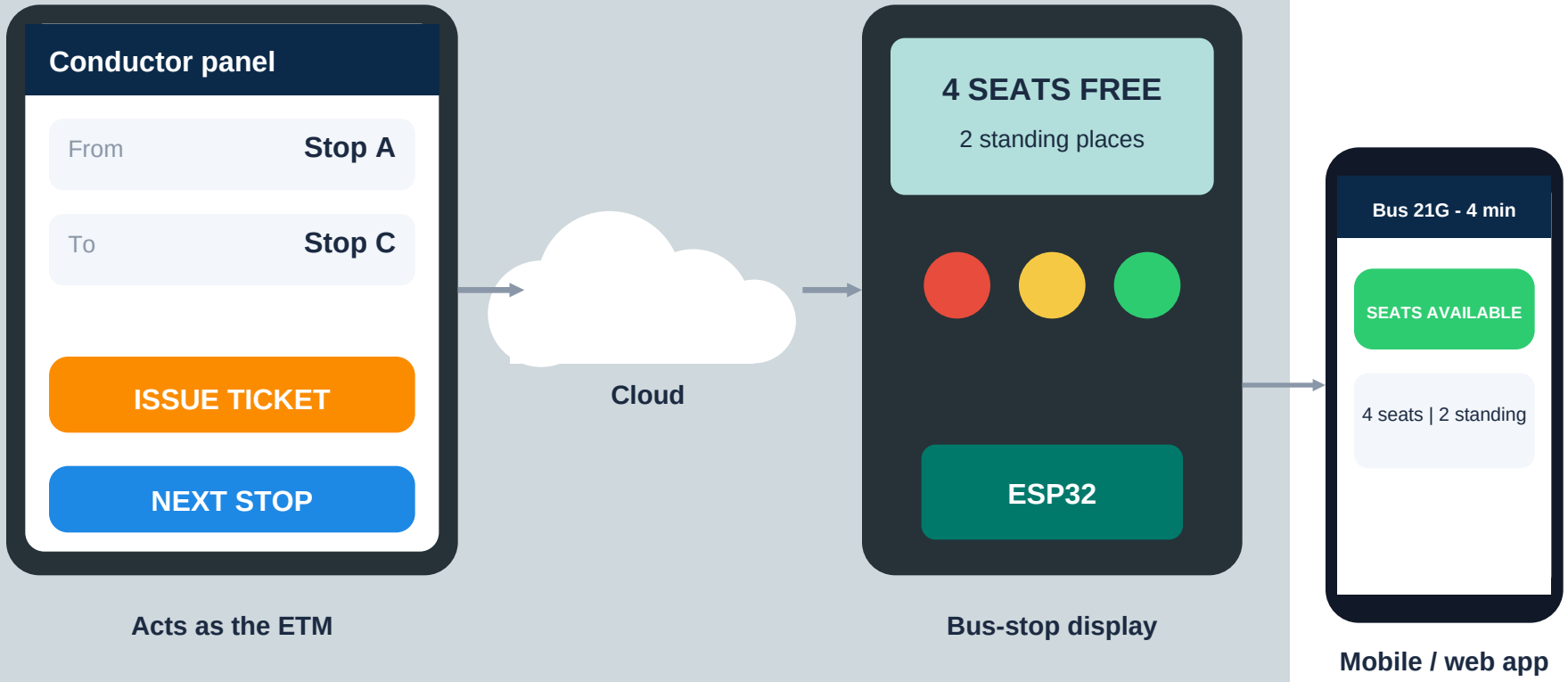
3 No-show protection

If the booked person does not board within 2 stops, the seat is released.

Working Prototype (Demo Model)

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Low-cost, easy to build



What it shows

- Panel issues tickets like the ETM
- Seats fill, then status turns Yellow
- "Next stop" frees seats automatically
- LCD, LEDs and phone update live
- Booking A to C frees the seat after C

Why It Is Different

Feature	Normal map apps	Live-tracking apps	SeatSense
Bus location and arrival time	Yes	Yes	Yes
Number of free seats	No	Limited	Yes
Standing space information	No	No	Yes
Seat booking for part of journey	No	No	Yes
Bus-stop display board	No	Limited	Yes
Uses the conductor's existing ETM	No	No	Yes

General comparison for presentation; some apps show crowd level in some cities. Our edge: seats + standing + segment booking.

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IDEALab

Know before you wait. Travel with confidence.

- Live seats and standing space for the next bus
- Green / Yellow / Red status on phone and bus-stop board
- Seat booking only for the part of the journey you travel
- Uses the conductor's ETM, scalable to city buses

Thank you - Questions?

